

Deterministic Finite Automaton (DFA)

A DFA reads input and transitions from one state to another according to the symbol read from the input.

If the input ends and the DFA is in an accepting (final) state, the input is accepted.

If the input ends and the DFA is in any state other than an accepting state, the input is not accepted.

A DFA is defined as

$Q = \{q_1, q_2, \dots, q_n\}$ a finite set of states

$\Sigma = \{a, b, c, d, \dots, z\}$ a finite set of input alphabets, can be any symbols

$\delta = Q \times \Sigma \rightarrow Q$ a finite set of state transitions

$s \in Q$ initial state

$F \subset Q$ (non-empty) set of accepting states

Properties of state transitions

$\delta(q, \lambda) = q$ where λ is the empty string

$\delta(q, ab) = \delta(\delta(q, a), b)$, chaining of transitions

Let $x = (x_1 x_2 \dots x_n)$ string, $x_i \in \Sigma$

The DFA accepts x if $\delta(q, x) \in F$ (ends in an accepting state)

and rejects it if $\delta(q, x) \notin F$ (ends in a non-accepting state)

Example 1.

Let's make a parking meter where you can insert coins of 10, 20, and 50. When the one euro parking fee is reached, the machine accepts the input and you can park. Yeah, this is pretty cheap.

What states are needed?

What kinds of transitions are needed?

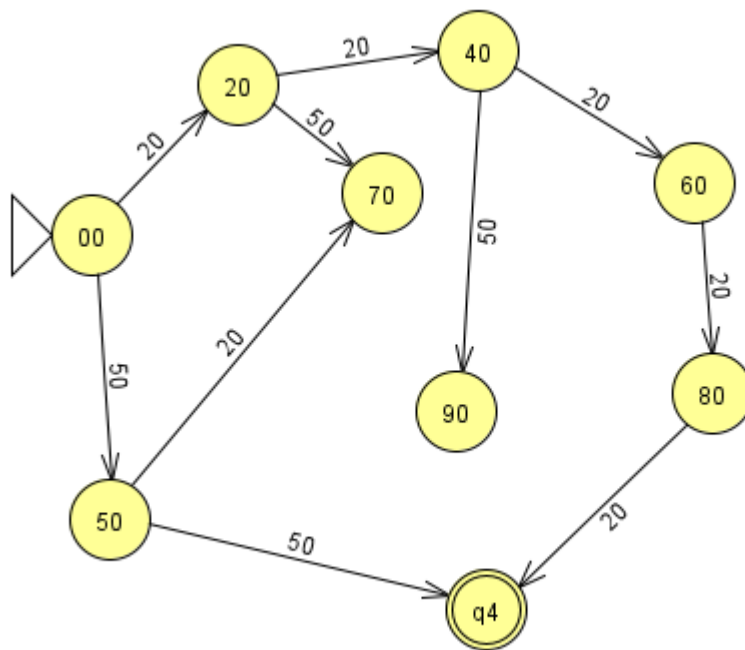
How to represent the coins (input)?

Are there multiple accepting states?

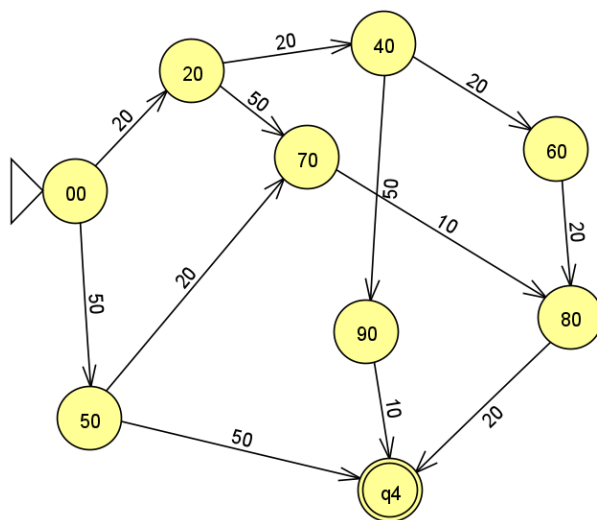
Let's tabulate the states and transitions:

Value	change 20	change 50
00	20	50
20	40	70
40	60	90
60	80	yli
80	100	yli

Based on this, draw an automaton (JFLAP) where first the coins are 20 and 50



OK, without ten cents it won't work:



Input	Result
2020202020	Accept
5050	Accept
50201020	Accept
20501020	Accept

Let's tabulate the states and transitions:

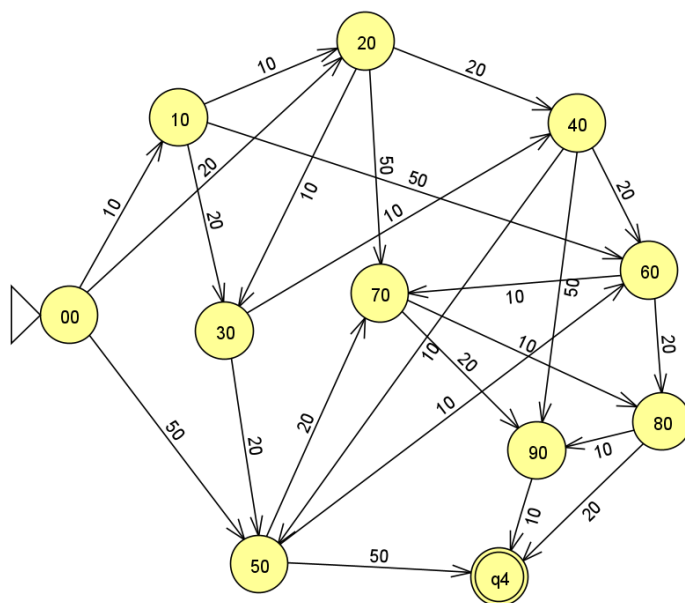
VALUE (STATE)	change 10	change 20	change 50
00	10	20	50
10	20	30	60
20	30	40	70
30	40	50	80
40	50	60	90
50	60	70	100
60	70	80	yli
70	80	90	yli
80	90	100	yli
90	100	yli	yli

Based on this, draw an automaton (JFLAP) where the coins are 10, 20 and 50

When adding 10 cents to the whole, more states are needed, what?

Example 2

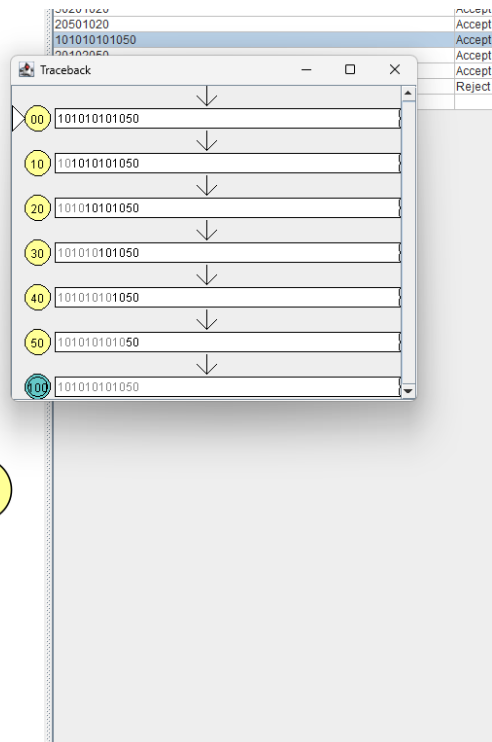
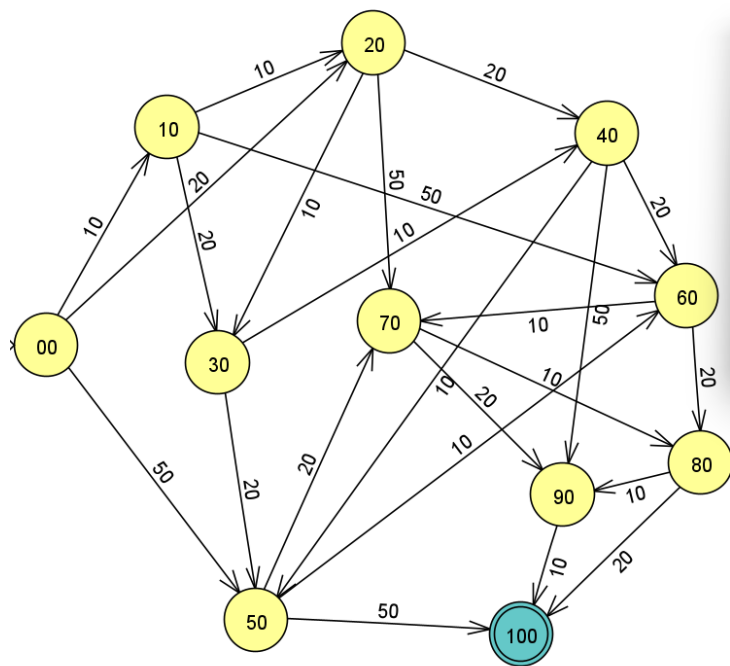
Including 10 cents in the automaton...



Input	
2020202020	Accept
5050	Accept
50201020	Accept
20501020	Accept
101010101050	Accept
20102050	Accept
5010101020	Accept

NOTE 1. Naming states as 70 tells more than q7

NOTE 2. You can get the automaton's "execution trace" from the Trace button



λ -transition

Final states could be minimized to one with an empty transition. When the input ends, if there is an empty transition from a state to a final state, the input is accepted. **However, then the automaton is no longer deterministic.**

Shortly about regular expressions

- * zero or more: a^* can be a , aa , aaa , $aaaa...a$ or empty
- + one or more: a^+ can be a , aa , aaa , $aaaaa...a$ but not empty
- ? One or none $a?$ is either a or empty
- Parentheses denote the scope of effect $(ab)^+$ can be ab , $abab$, $ababab...ab$